1. Project Name

FoodWise: AI-Powered Expiry-Aware Recipe Recommendation for Sustainable Food Consumption (SDG 12.3).

Github: https://github.com/duhaozyakup/Innovation-Campus-Assignment---Group5---FoodWise/tree/main

2. Overview

Household and institutional food waste is a significant environmental and economic issue. The United Nations Environment Programme (UNEP) estimates that over 1.05 billion tonnes of food were wasted globally in 2022; households accounted for 60% and food-service operations for 28%. FoodWise focuses on helping households and community kitchens make better use of ingredients they already have before they spoil.

FoodWise is designed as an inventory-aware recipe recommendation project. The planned user workflow allows people to enter ingredients, quantities, and use-by dates; the local Python prototype currently provides recipe retrieval and an initial urgency-aware ranking method. The system aims to support more efficient food use and aligns with the goals of SDG 12.3; its effect on food waste has not been measured. Its potential contribution to SDG 2 is indirect.

User Segment	Anticipated context (not yet user-validated)	Proposed FoodWise value
Urban Households, Students, and Working Professionals	Potential household challenge: manual pantry entry and meal planning can be inconvenient, and perishable food may be forgotten. These assumptions need user research.	The current prototype ranks recipes using available ingredients and user-entered dates to highlight urgent stock. A future barcode lookup could fetch a product profile for user confirmation; it does not read the individual package date. Turkish cuisine and ingredient matching are EDA priorities, but cross-language matching is not yet validated.
Community Kitchens, Food Pantries, and Student Co-ops	Potential community-kitchen challenge: donation quantities and timing may vary, while menus and available records can limit planning. This scenario needs validation with a partner.	Future extension: batch inventory entry, multi-serving recipe scaling, and recommendations for short-dated donations. These capabilities require partner data and testing; they are not established by the current prototype.

3. Background

Natural Language Processing (NLP) and Information Retrieval (IR) provide methods for ingredient-based recipe search. In the processed English EDA samples, salt appears in 38.02% of the Food.com sample and 43.00% of the bounded RecipeNLG sample; other common ingredients also vary by source. This motivates comparing simple ingredient overlap with TF-IDF weighting, but frequency statistics alone do not show which ranking method works best. The local prototype implements Jaccard overlap, TF-IDF cosine similarity, and optional latent semantic analysis (LSA). Transformer-based embeddings and synonym resolution are possible future approaches, particularly for Turkish ingredient matching. An expiry-aware ranking can combine recipe relevance, ingredient coverage, and user-entered urgency while showing missing ingredients.

4. Key Objectives / Business Objectives

Develop and evaluate a practical recipe-discovery workflow that combines pantry ingredients, recipe relevance, and user-entered urgency; adapt its data preparation for Turkish recipes; and identify what evidence is still needed before barcode, food-bank, dietary, or waste-impact claims can be made.

a. Research Questions

- 1. How do available English and Turkish recipe datasets differ in composition and data quality, and what preprocessing is needed for Turkish-focused ingredient matching?

- 2. How well does an illustrative pantry inventory match recipes using normalized ingredient overlap, and what should be evaluated when comparing ingredient-weighting methods?
- 3. How could expiry-aware, barcode-assisted pantry management prioritize recipes and support measurable waste reduction while respecting data and safety limits?

b. Key Steps

- Prepare and profile English and Turkish recipe sources, FoodKeeper storage references, and the fields needed for a future barcode-assisted inventory.
- Normalize ingredient text and quantities while keeping Turkish terms distinct until aliases and translations are validated; review duplicate candidates rather than merging them automatically.
- Build recipe retrieval baselines and an expiry-aware ranking prototype; expose ingredient coverage and missing ingredients, and compare methods with suitable relevance tests.
- Design a pantry-entry workflow with manual input and a proposed barcode-to-Open-Food-Facts lookup, and keep product identity, package dates, and FoodKeeper storage guidance separate.
- Plan user and community-kitchen evaluation, including baseline/post-use waste measurement, and add dietary filters only if validated nutrition data and qualified review become available.

5. Methods and Workflow

a. Datasets

The processed-data run covers 2,242,462 recipe records overall. Detailed analysis is bounded to 5,000 records per non-Turkish source; all 3,320 available Turkish recipes are retained for Turkish-specific profiling.

Dataset / source	Records and role	Scope / caveat
Food.com sample	8,000 English recipes	Reproducible curated sample; used for the illustrative pantry-overlap calculation.
RecipeNLG	2,231,142 English recipes in processed corpus	5,000-record bounded EDA sample; bundled license limits use to non-commercial research and education.
Turkish recipes	3,320 recipes	All retained for detailed profiling; no dataset-wide license is recorded in the local source.
FoodKeeper	951 populated rows; 661 Product rules	Separate storage-reference data; 647 Product rules contain at least one numeric duration.
Open Food Facts	No records/API extract in this project run	Proposed barcode product lookup only; no Turkish barcode coverage or product-field completeness was measured.
Municipal kitchen / food-bank records	No partner inventory, donation, adoption, or discard logs	Future pilot evidence: permissioned, de-identified dated quantities received, used, donated, and discarded; discard reasons; recommendations shown/adopted.
UNEP Food Waste Index 2024	External context; reported 102 kg/person/year household-waste estimate for Türkiye	Used only for hypothetical scenarios, not a FoodWise result. Verify the country estimate against the source table before external use.
Turkish nutrition / allergen data	No validated Turkish nutrition, allergen, or diet-suitability fields	Needed before diet-specific or diabetes-related recommendations; requires validated data and qualified professional review.

b. Data Cleaning and Preprocessing

- Stream the large RecipeNLG source and retain source identity; use the reproducible 8,000-row Food.com sample and the full available Turkish collection for detailed Turkish profiling.
- Normalize ingredient strings with Unicode NFKC, case folding, whitespace collapse, and Turkish dotted/dotless-I handling. Keep English and Turkish ingredient terms separate; do not infer translations, synonyms, or safe substitutions.
- Derive ingredient counts, source-specific term frequencies and IDF values, Turkish category/cooking-method summaries, quantity/unit completeness, and exact-string inventory coverage.

- Audit repeated Turkish ingredient sets, normalization variants, and Jaccard near-duplicate candidates as review signals only. Preserve FoodKeeper native duration units and text bounds; approximate day conversions are descriptive, not safety rules.

Key EDA results: the Turkish collection has 1,276 normalized ingredient terms, a median of 9 ingredients and 11 instruction steps; amounts occur on 82.3% of ingredient entries and units on 78.0%. The 480 Turkish near-duplicate pairs at Jaccard ≥ 0.80 are candidates, not approved merges.

c. Modelling

Project workflow: pantry entry, ingredient normalization, recipe candidate retrieval, coverage and urgency scoring, ranked suggestions with missing ingredients shown. Barcode lookup is a future input path; the current local prototype uses recipe and inventory records, not an Open Food Facts connection.

The project retrieval prototype in src/foodwise.py provides Jaccard ingredient overlap, TF-IDF cosine similarity, and optional 64-dimensional latent semantic analysis (LSA). Its synthetic source-recipe recovery benchmark tests whether a recipe can be retrieved from a subset of its own ingredients; it does not measure user relevance. Separately, the illustrative nine-item inventory matches at least 25% of ingredient names in 1,502 of 8,000 Food.com recipes (18.77%); 119 (1.49%) reach 50% coverage, and one has complete lexical coverage. This is not an observed household inventory or a Turkish/community-kitchen result. Real inventory data retrieved from local kitchens will be incorporated in a future evaluation.

The local recommender ranks recipes using a transparent initial heuristic: 0.60 TF-IDF cosine similarity, 0.30 ingredient coverage, and 0.10 coverage of urgent ingredients. Inventory items with user-entered dates within two days are treated as urgent; past-due or explicitly unusable items are excluded. Recipes containing all known recipe ingredients are sorted ahead of incomplete matches, and missing ingredients are shown rather than hidden. These weights are prototype settings, not learned or user-validated parameters; quantities, substitutions, nutrition, and food safety are not inferred.

For the complete product workflow, users can add items manually or, in a future version, scan a barcode to request an Open Food Facts product profile. The user must confirm or correct the match, enter quantity, and separately enter or verify any package date; an unknown or ambiguous lookup falls back to manual entry. This API integration has not been built or evaluated, and its community-contributed records may be incomplete or incorrect [1].

The project's sustainability outcome must ultimately be assessed with measured use and waste data. The EDA includes only an illustrative scenario based on a reported 102 kg/person/year Türkiye baseline attributed to UNEP [2]: a four-person household implies 408 kg/year, and hypothetical 1%, 5%, and 10% reductions equal 4.08, 20.40, and 40.80 kg/year. These are arithmetic scenarios, not FoodWise results; the source-country estimate should be checked against the report table before external use.

FoodKeeper is currently profiled as a separate reference dataset; product-category-to-ingredient matching has not been built or validated, so its storage ranges are not used to invent missing package dates. The project data also contain no Open Food Facts records, municipal partner logs, measured household waste, or validated Turkish nutrition/allergen fields. A community pilot would require permissioned, de-identified quantities received, used, donated, and discarded; medical-diet recommendations require validated nutrition data and qualified professional review.

d. Deliverables

- Project data-preparation pipeline, reproducible source manifests, and executed notebooks: notebooks/FoodWise_Assignment1.ipynb and notebooks/FoodWise_Processed_Data_EDA.ipynb.
- Python retrieval prototype and comparison methods: src/foodwise.py (Jaccard, TF-IDF, optional LSA, and an initial expiry-aware scoring heuristic).
- EDA tables, figures, inventory coverage, research-question evidence, and data-readiness outputs: outputs/processed_eda/; report: report/FoodWise_Assignment1_EDA_Report.docx.
- Next project milestones: evaluate ranking with judged relevance and diverse pantry cases; validate Turkish ingredient aliases; assess barcode coverage and API reliability; test usability with households or a community partner; and collect comparable baseline/post-use waste measurements before claiming impact.

Sources: [1] Open Food Facts API documentation, https://openfoodfacts.github.io/openfoodfacts-server/api/. [2] UNEP, Food Waste Index Report 2024, https://www.unep.org/resources/publication/food-waste-index-report-2024. [3] RecipeNLG, https://recipenlg.cs.put.poznan.pl/. Food.com provenance: https://www.kaggle.com/datasets/irkaal/foodcom-recipes-and-reviews. Run details: processed_data/runs/20261022T132846Z-b7e73fc9/run_manifest.json; EDA: outputs/processed_eda/analysis_manifest.json.

Group 5
Nisa Sunar
Emirhan Tan
Erdem Açıkel
Duha Özyakup